

$$1. \text{Cat}(h_{i-1}, x_i) = [2, 3]$$

$$\begin{aligned} 2. \sigma_f([2, 1]) &= \text{act}(2*0 + 3*0.6) \\ &= \text{act}(1.8) \\ &= 1 \end{aligned}$$

$$\begin{aligned} 3. \sigma_{in}([2, 1]) &= \text{act}(2*0.5 + 3*0.5) \\ &= \text{act}(2.5) \\ &= 2 \end{aligned}$$

$$\begin{aligned} 4. \sigma_o([2, 1]) &= \text{act}(2*0.25 + 3*0.5) \\ &= \text{act}(2) \\ &= 2 \end{aligned}$$

$$\begin{aligned} 5. c_{i-1} * \sigma_f([2, 1]) &= 1 * 1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} 6. 1 + \sigma_{in}([2, 1]) &= 1 + 2 \\ &= 3 \quad \Rightarrow c_i \end{aligned}$$

$$7. 3 * \sigma_o([2, 1]) = 6 \quad \Rightarrow o_i, h_i$$